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Areas of physics by complexity



Newton's
Mechanics

Electro-
Magnetism

Special
Relativity

Quantum Mechanics
General Relativity

Quantum
Field Theory

Complexity
Science

Project # 37: Explosive Percolation

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Phase transitions are defined as an abrupt change in the some properties of a system as a result of a little change in the controlling variables, change that could happen in a discontinuous way (first-order transition) or continuous (second-order transition). In the following it will be presented numerical proofs that advocate that percolation phase transition can in fact happen in a discontinuous way in a random network as a product of particular selection rules used in the process. We consider percolation phase transition in models of random network formation, comparing the results obtained in:

- Erdős–Rényi model (ER): a network formation model where, starting from an unconnected network, at each step two random nodes are picked uniformly at random and connected together.
- Bohman-Frieze model: a model of random network formation that relies on "bounded-size" selection rule (meaning a selection rule that resolves a dichotomy between two edges only by the sizes of the component at the four end-points of the edges), that selects the first edges from the two sampled only if all the nodes attached to it as size at maximum K (picked $K=1$), otherwise chooses the other edge.
- Product-rule model: a model based on a specific "unbounded-size" selection rule. This selection rule consists in choosing between two edges by analyzing the product of the sizes of the connected components an edge joins and choosing the edge that minimizes this product.

With the latter, that initially aims to slow the emergence of the largest connected component, there will be associated numerical proofs of a discontinuous phase transition [2](while it will be shown that the same type of rule used to speed up the emergence of the largest connected component leads to a normal continuous transition). Since there are already numerical and mathematical considerations that discards the idea that a discontinuous phase transition can happen with a "bounded-size" selection rule, numerical proves will be brought to support the idea that "unbounded-size" selection rule can give rise to a discontinuous phase transition. The discontinuity of the process is shown by analyzing the ratio between the size of the largest connected component (references as C) and the number of total vertices as a function of the number of edges. It will be considered the interval of time-steps $\Delta = t_1 - t_0$ as the difference between the last time-step at which $C < n^\gamma$ and the first time-step at which $C > An$, with $A > 0$ and $\gamma \in (0, 1)$. While in continuous transition this interval scales linearly with the size of the network, it will be shown that with PR the transition will scale as $\Delta \propto n^\beta$ with $\beta \in (0, 1)$, advocating for a discontinuous phase transition in the limit of $n \rightarrow \infty$. Even though this result will be shown for $A = 0.5$, $\gamma = 0.5$ with a scaling of $\Delta < 2n^{2/3}$ in the product rule case, it will be later shown that this holds generally for the variables previously shown, with a relation of $\gamma + \lambda\beta = \mu$ with $\lambda \approx 1.2$ and $\mu \approx 1.3$.

Against these proofs there is a counter-argument in the article "*Explosive Percolation is Continuous*" [4] that shows mathematically by contradiction that the proofs brought were wrong. It's important to remark that this proof is specific to the Achlioptas processes, where starting from n isolated nodes a certain number of potential edges are sampled and chosen following a specific rule. In fact, in processes of different nature the first order transition is observed, like in the paper "*Sudden Emergence of a Giant k -Core in a Random Graph*" by Boris Pittel, Joel Spencer and Nicholas Wormald [1].

In Fig. 5 it has been shown the evolution of the ratio C/n as a function of n for the three model taken into consideration. As it can be seen, while we can see a very clear trend for the ER and BF model, the PR model shows an abrupt transition.

In the following it will be shown that while ER and BF models do indeed come across a phase transition (with C scaling as $\log(n)$ for $r < 1/2$ and $C \approx (4r - 2)n$ for the ER model), the transition it is indeed continuous. It will be also shown that while the PR rule reveals the previously anticipated "discontinuous" behavior when the rule tries to minimize the emergence of the LCC, in the opposite case the transition is perfectly continuous. It will be then brought to attention how the atypical behavior observed is not strictly related to the PR rule but shows even with different models.

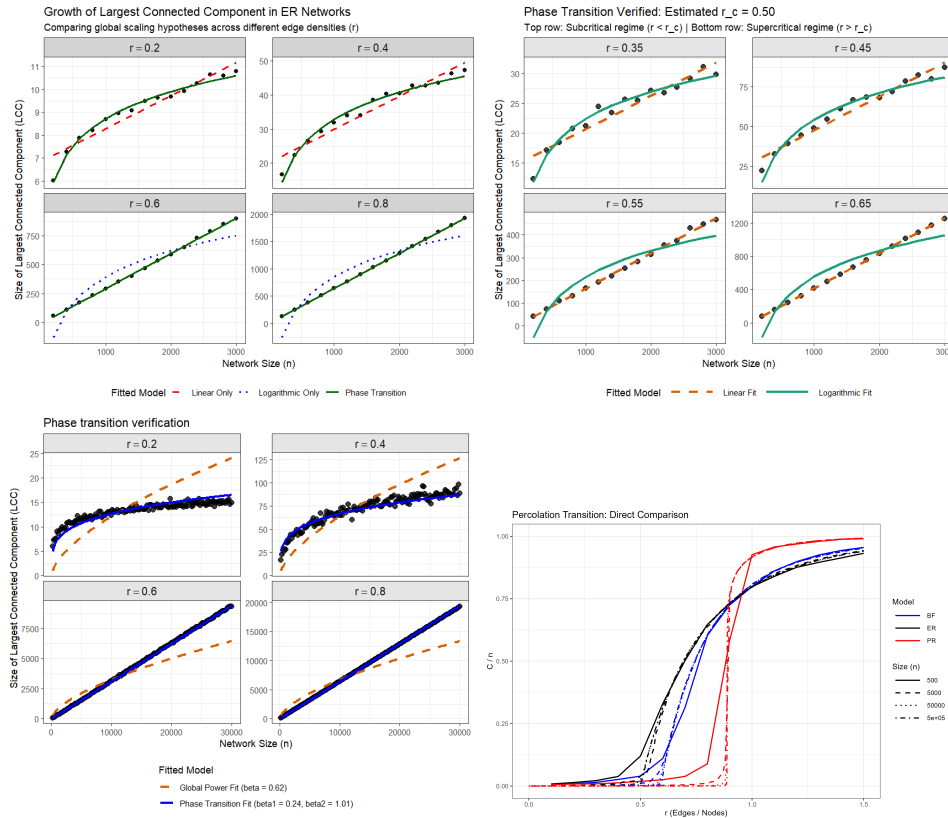


Figure 1: On top left: fitting of size of LCC with ER model for $r = 0.2, 0.4, 0.6, 0.8$ as a function of n . The model by which the fitting has been done are a completely logarithmic model, a completely linear model and a model that is logarithmic for $r < 0.5$ and linear for $r > 0.5$ that should capture a phase transition. For each n used for the analysis we average the values taken from 300 synthetical network to reduce stochastic fluctuations effects.

On top right: fitting of size of LCC with BF model for $r = 0.2, 0.4, 0.6, 0.8$ as a function of n , also using a maximum likelihood approach to find the phase transition point. The scaling of the BF model confirms also the results in the paper "The Bohman-Frieze Process Near Criticality" [3].

On the bottom left: fitting of size of LCC with PR model (picking the choice that doesn't slow the emergence). Without knowledge of the exact scaling before and after the transition it has been performed a polynomial fitting in logscale to show that the trend before and after $r = 0.5$ is significantly different, with an exponent of $\beta = 0.287$ before and $\beta = 1.015$ after and a loglikelihood ratio of $LLR = 114$ with respect to the hypothesis of no phase transition.

On the bottom right: Scaling of the size of the LCC (normalized) as a function of the ratio between the number of edges and size of the network for network of different size and for the ER, BF and PR model.

These plots and fitting brings conclusive evidence of the presence of a phase-transition in the case in exam. As for the scaling of Δ with respect to n :

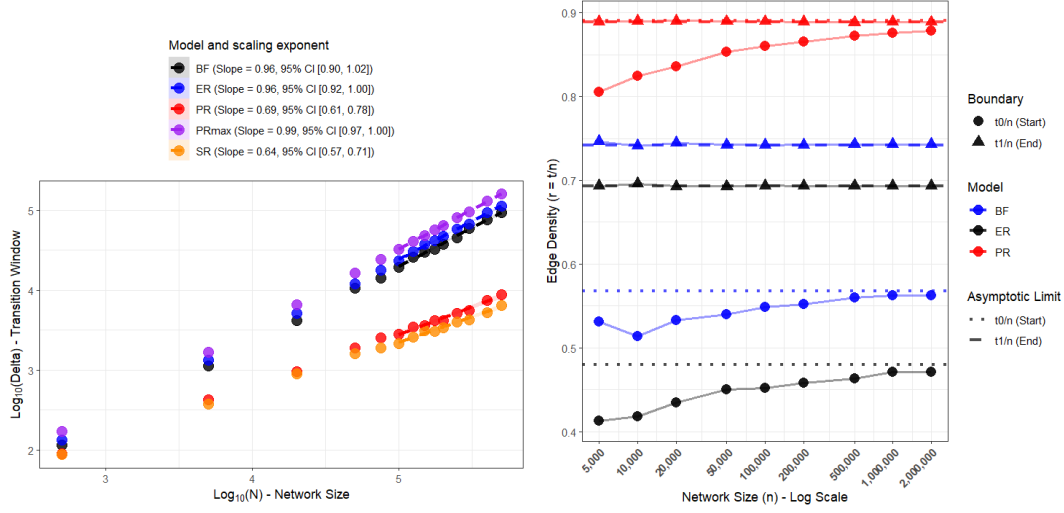


Figure 2: **On the left:** fitting of scaling of $\Delta = t_1 - t_0$, the difference between the last time-step at which $C < n^{1/2}$ and the first time-step at which $C > 0.5n$ as a function of n . With a model of $C \propto kn^\alpha$ we obtain for the ER model an $\alpha_{ER} = 0.977$ with credible interval $[0.941, 1.01]$, for the BF model an $\alpha_{BF} = 0.958$ with credible interval $[0.918, 0.999]$ and for the PR model (choosing the edges in order to slow the emergence of the LCC) an $\alpha_{PR} = 0.673$ with credible interval $[0.602, 0.641]$ **On the right:** plot of the values of t_0/n and t_1/n as a function of the size of the network. It can be noticed that in the case of the PR model the two points tend to converge to a common value $r_c = 0.88826$ with a 95% credible interval of $[0.88562, 0.89090]$ with a maximum likelihood approach and a $r = c + n^\beta$ model.

Even though the fitting for the ER and BF model does not show a slope exactly equal to $\alpha = 1$ it supposedly is due to the fact that we are not analyzing large enough networks (more evidence of this is brought in the additional materials). The scaling of the PR and the Sum Rule (SR) model is on the other hand undoubtedly not linear, with values of $\alpha \approx 0.6$, that is in line with the scaling of $n^{2/3}$ previously anticipated. Also we can see that the PR rule that boosts the emergence of the LCC presents a scaling comparable to the bounded-size rule, suggesting a second order phase-transition.

To confirm that this results are not strict to the values of $A = 0.5$ and $\gamma = 0.5$ we perform a fit of $\Delta_{A,\gamma}$ to show that the relation is more general and follows the rule $\gamma + \lambda\beta = \mu$, where $\beta < 1$ represents the scaling of the interval Δ with respect to n .

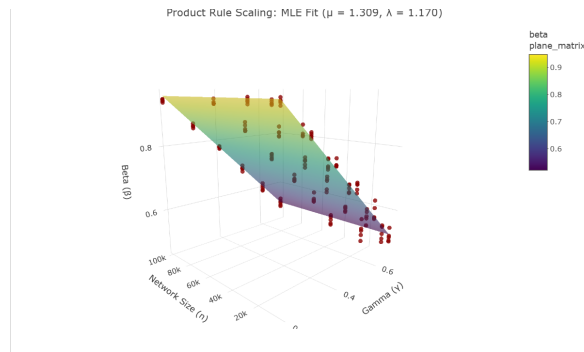


Figure 3: comparison of the scaling of $\Delta_{A,\gamma}$ as a function of n with hypothesis $\Delta \propto n^\beta$. The comparison shows perfect correspondence with the model $\gamma + \lambda\beta = \mu$ with $\lambda \approx 1.2$ and $\mu \approx 1.3$ as anticipated. While a point estimation of β is performed, the parameters μ and λ are obtained with a maximum likelihood fit.

0.1 | Appendix

- **Finite size effects:** as the values of the fitting of the bounded size rules are not exactly linear in n or constant depending on the case, a following analysis has been done to understand whether the results were incorrect or if they were only due to the finite size of the network, the results are the following:

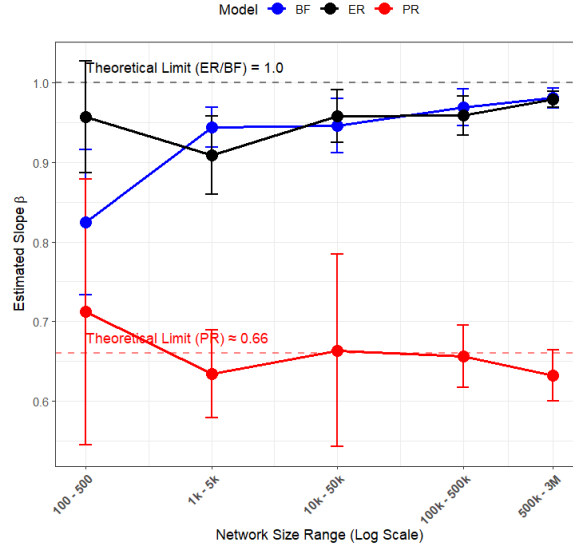


Figure 4: Fitting of the scaling of Δ as a function of n for different ranges of n used with a maximum likelihood estimate with 95% error bars. This analysis has the goal to justify that the analysis reported didn't exactly match the theoretical expectation due to a finite size effect.

- **Union-find simulation:** since the results we are looking for in the analysis are based on network near the thermodynamic limit and due to low computational resources optimization is key to have enough proof to reproduce the results. As a result, the *igraph* library could not offer such requirement and so a Union-Find simulation was needed to obtain satisfying results. To achieve the computational efficiency necessary for simulating macroscopic networks (of the order of $N = 10^6$ nodes), the simulation employs a highly optimized data structure. Standard graph traversal algorithms require recalculating the connected components at every step. In contrast, the Union-Find algorithm dynamically maintains the partitioning of the network's connected components as edges are introduced, reducing the computational complexity to almost constant. The structure is initialized as a forest of N disjoint trees, where each vertex serves as its own root and represents an isolated component of size one.

With Achlioptas processes, the algorithm must rapidly determine the sizes of the components those edges would connect. This is achieved using a Find operation equipped with a path compression (make each point refer to a root point instead of reconstructing each time the connected components). By retrieving the component sizes saved directly at the root nodes, the algorithm can instantaneously evaluate the selection rule criteria. Once the optimal edge is chosen, the components are merged using a union by size, which attaches the root of the smaller component to the root of the larger one.

- **Averaging values** : As fluctuation could lead to inaccurate conclusion, the plotted data is obtained by taking the average of the value for each time-steps of independent realizations of the same model
- **Asymptotic values of t_0/n and t_1/n for product rule**: By fitting with a joint NLS fit the boundaries with a model of $c + n^\beta$ we obtain an asymptotic value for r_c in the Product Rule model of 0.8883. The results are the following:

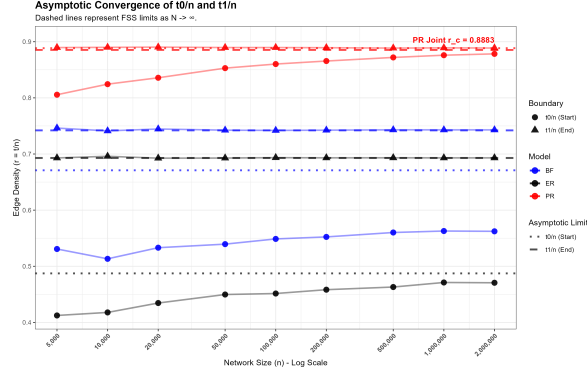


Figure 5: fitting with a joint NLS fit the boundaries t_0/n and t_1/n of the PR model with an hypothesis of $c + n^\beta$.

1 | Bibliography

- [1] Nicholas Wormald Boris Pittel, Joel Spencer. Sudden emergence of a giant k -core in a random graph. 2011.
- [2] Joel Spencer Dimitris Achlioptas, Raissa M. D'Souza. Explosive percolation in random networks. *Science*, 2009.
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- [4] Lutz Warnke Oliver Riordan. Explosive percolation is continuous. *Science*, 2011.